1 $a_z(\bar{h})$ from the plane $c_\perp$

(bars denote lengths divided by R: $\bar{h} := h/R$, sphere centre to wall surface, contact at $\bar{h} = 1$; $a_{\text{nom}} := \Delta t/\gamma_N$ [μm/pN]; $\bar{a}_z := a_z/a_{\text{nom}}$)

$$c_\perp(\bar{h}) = \frac{1}{1 - \frac{9}{8}\frac{1}{\bar{h}} + \frac{1}{2}\frac{1}{\bar{h}^3} - \frac{57}{100}\frac{1}{\bar{h}^4} + \frac{1}{5}\frac{1}{\bar{h}^5} + \frac{7}{200}\frac{1}{\bar{h}^{11}} - \frac{1}{25}\frac{1}{\bar{h}^{12}}}$$

$$a_z(\bar{h}) = \frac{a_{\text{nom}}}{c_\perp(\bar{h})}$$

$$\bar{a}_z(\bar{h}) = 1 - \frac{9}{8}\frac{1}{\bar{h}} + \frac{1}{2}\frac{1}{\bar{h}^3} - \frac{57}{100}\frac{1}{\bar{h}^4} + \frac{1}{5}\frac{1}{\bar{h}^5} + \frac{7}{200}\frac{1}{\bar{h}^{11}} - \frac{1}{25}\frac{1}{\bar{h}^{12}}$$

$$\bar{a}_z(1) = 0, \qquad \left.\frac{d\bar{a}_z}{d\bar{h}}\right|_1 = 1, \qquad \lim_{\bar{h}\to\infty} \bar{h}\left(1 - \bar{a}_z\right) = \frac{9}{8}$$

2 Height coordinate and the wall surface

($\bar{w} := w/R$, particle centre along the wall normal; $\bar{w}_s$ = position of the wall surface on the same axis, known; the plane's surface is the origin of $\bar{h}$, so $\bar{w}_s = 0$ there)

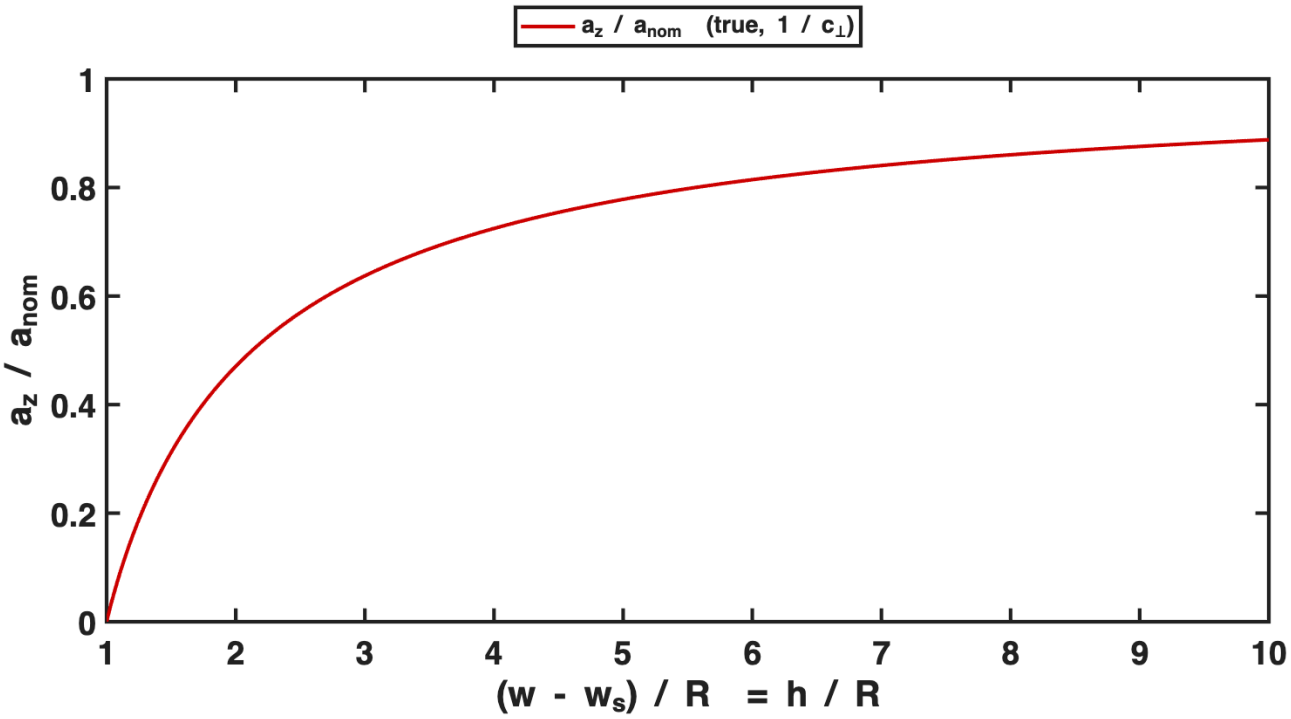
$$\bar{h} = \bar{w} - \bar{w}_s$$

contact: $\bar{w} - \bar{w}_s = 1$

band used: $\bar{w} - \bar{w}_s \geq 1.1$

3 a_z as a function of $\bar{w} - \bar{w}_s$

$$\bar{a}_z(\bar{w} - \bar{w}_s) = 1 - \frac{9}{8}\frac{1}{(\bar{w} - \bar{w}_s)} + \frac{1}{2}\frac{1}{(\bar{w} - \bar{w}_s)^3} - \frac{57}{100}\frac{1}{(\bar{w} - \bar{w}_s)^4} + \frac{1}{5}\frac{1}{(\bar{w} - \bar{w}_s)^5} + \frac{7}{200}\frac{1}{(\bar{w} - \bar{w}_s)^{11}} - \frac{1}{25}\frac{1}{(\bar{w} - \bar{w}_s)^{12}}$$



4 a_z with the surface gap scaled by s

(s scales the gap $(\bar{w}-\bar{w}_s)-1$ between the sphere surface and the wall; $s=1$ is the plane)

$$\bar{a}_z(\bar{w}-\bar{w}_s) = 1 - \frac{9}{8}\frac{1}{1+s\left((\bar{w}-\bar{w}_s)-1\right)} + \frac{1}{2}\frac{1}{\left(1+s\left((\bar{w}-\bar{w}_s)-1\right)\right)^3} - \frac{57}{100}\frac{1}{\left(1+s\left((\bar{w}-\bar{w}_s)-1\right)\right)^4} + \frac{1}{5}\frac{1}{\left(1+s\left((\bar{w}-\bar{w}_s)-1\right)\right)^5} + \frac{7}{200}\frac{1}{\left(1+s\left((\bar{w}-\bar{w}_s)-1\right)\right)^{11}} - \frac{1}{25}\frac{1}{\left(1+s\left((\bar{w}-\bar{w}_s)-1\right)\right)^{12}}$$

(δ shifts the effective wall below the reference surface; $\delta=0$ is the plane; band $1.1 \le \bar{w}-\bar{w}_s \le 5.5$, least squares on the Fig. 11 curves, probe_plane_offset_law.m)

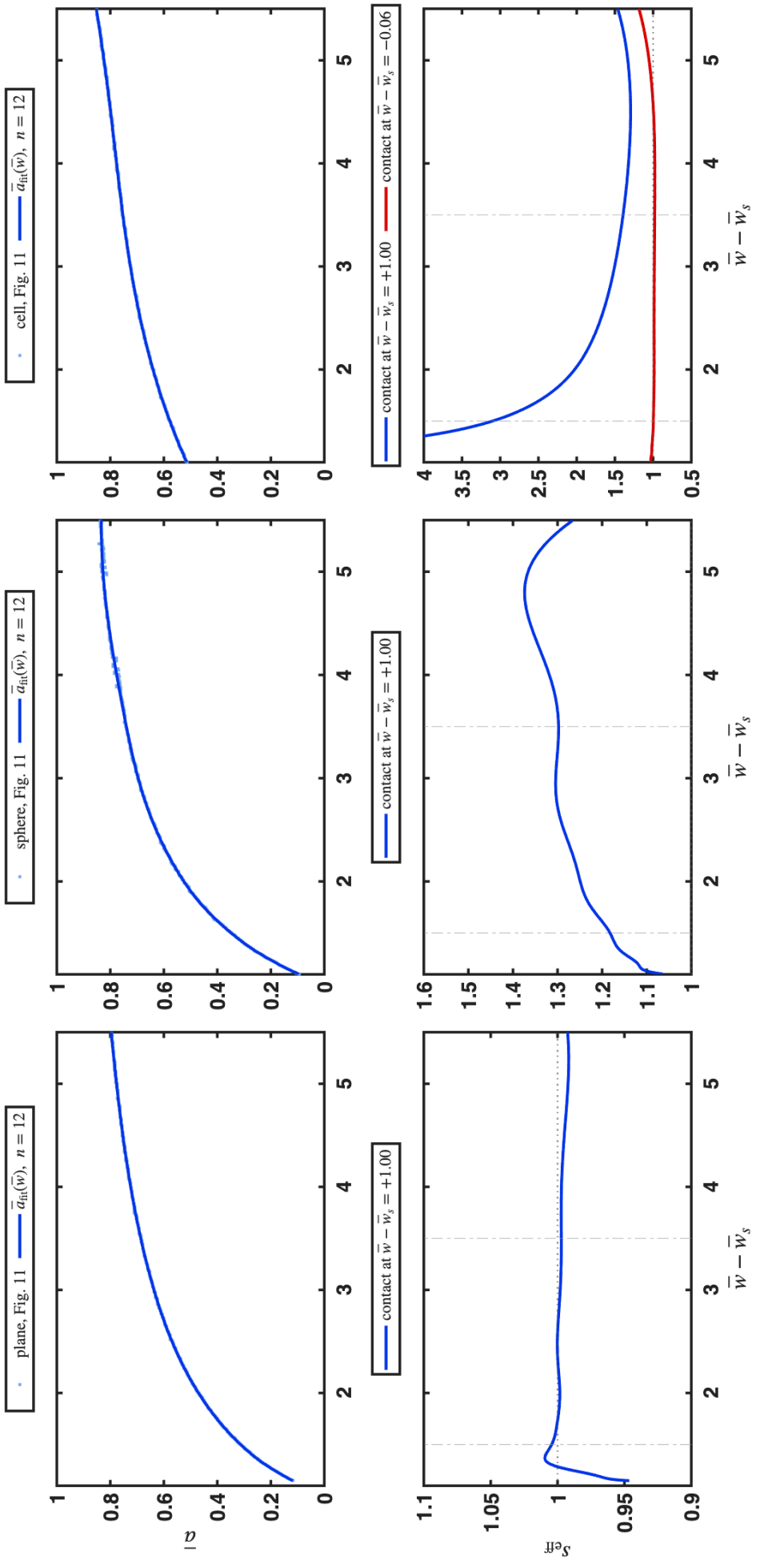
$$\bar{a}_z(\bar{w}-\bar{w}_s) = D_\perp\Big(\frac{1}{(\bar{w}-\bar{w}_s)+\delta}\Big)$$

$$\bar{a}_z(\bar{w}-\bar{w}_s) = D_\perp\Big(\frac{1}{1+s\left((\bar{w}-\bar{w}_s)+\delta-1\right)}\Big)$$

plane: $\delta=-0.001$ (rms 0.0016)

cell: $\delta=+1.050$ (rms 0.0061, 2.6%; with s free: $\delta=1.042$, $s=1.004$)

sphere: $\delta=-0.039$, $s=1.323$ (rms 0.0055, 20%; δ alone: rms 0.040)


plane, Fig. 11 — $\bar{a}_{\mathrm{fit}}(\bar{w})$, $n = 12$
sphere, Fig. 11 — $\bar{a}_{\mathrm{fit}}(\bar{w})$, $n = 12$
cell, Fig. 11 — $\bar{a}_{\mathrm{fit}}(\bar{w})$, $n = 12$
$\bar{a}$
contact at $\bar{w} - \bar{w}_s = +1.00$
contact at $\bar{w} - \bar{w}_s = -0.06$
s_{eff}
$\bar{w} - \bar{w}_s$

5 Two-point Padé form of $\bar{a}_z(\bar{w}-\bar{w}_s)$

($\bar{a}_z(\infty)=1$ built in; the two ends are the contact $\bar{w}-\bar{w}_s=1$ and the far field)

$$\bar{a}_z(\bar{w}-\bar{w}_s)=\frac{(\bar{w}-\bar{w}_s)^2+c_1\,(\bar{w}-\bar{w}_s)+c_0}{(\bar{w}-\bar{w}_s)^2+d_1\,(\bar{w}-\bar{w}_s)+d_2}$$

$$\bar{a}_z(1)=\frac{1+c_1+c_0}{1+d_1+d_2}$$

$$\frac{d\bar{a}_z}{d(\bar{w}-\bar{w}_s)}\Big|_1=\frac{(2+c_1)(1+d_1+d_2)-(1+c_1+c_0)(2+d_1)}{(1+d_1+d_2)^2}$$

$$\lim_{\bar{w}-\bar{w}_s\to\infty}(\bar{w}-\bar{w}_s)\left(1-\bar{a}_z\right)=d_1-c_1$$

$[1/1],\ c_1=d_2=0:\qquad \bar{a}_z(\bar{w}-\bar{w}_s)=\dfrac{(\bar{w}-\bar{w}_s)+c_0}{(\bar{w}-\bar{w}_s)+d_1}$

$$\bar{a}_z(1)=\frac{1+c_0}{1+d_1}$$

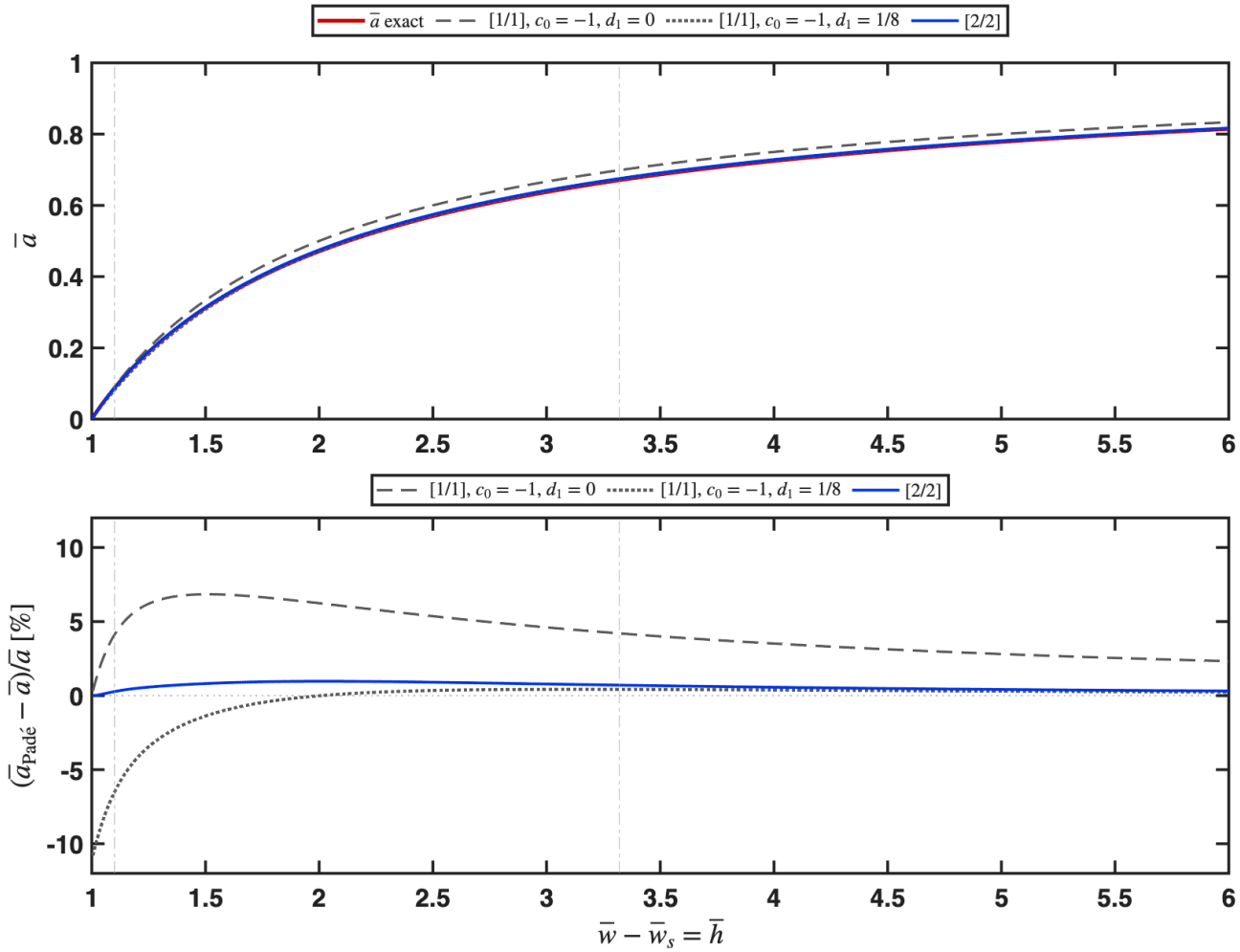
$$\frac{d\bar{a}_z}{d(\bar{w}-\bar{w}_s)}\Big|_1=\frac{d_1-c_0}{(1+d_1)^2}$$

$$\lim_{\bar{w}-\bar{w}_s\to\infty}(\bar{w}-\bar{w}_s)\,(1-\bar{a}_z)=d_1-c_0$$

plane, $\bar{a}_z(1)=0\ \Rightarrow\ c_0=-1:\qquad \bar{a}_z(\bar{w}-\bar{w}_s)=\dfrac{(\bar{w}-\bar{w}_s)-1}{(\bar{w}-\bar{w}_s)+d_1}$

$$d_1=0\ \text{ (slope 1)}\qquad\text{or}\qquad d_1=\frac{1}{8}\ \text{ (far }\tfrac{9}{8}\text{)}$$

$[2/2]$, plane (slope 1, curvature $-\frac{327}{200}$, far $\frac{9}{8}$): $\qquad c_0=\dfrac{102}{127},\quad c_1=-\dfrac{229}{127},\quad d_1=-\dfrac{689}{1016},\quad d_2=-\dfrac{1}{8}$


$\bar{a}$ exact
[1/1], $c_0 = -1$, $d_1 = 0$
[1/1], $c_0 = -1$, $d_1 = 1/8$
[2/2]
$\bar{a}$
$(\bar{a}_{\text{Padé}} - \bar{a})/\bar{a}$ [%]
$\bar{w} - \bar{w}_s = \bar{h}$

6 Two-point Padé form of the cell (Fig. 11), $\bar{w}_s$ = visible cell surface

($\bar{a}_z(\infty) = 1$ built in; $\bar{w}_s$ is the visible surface, the zero of the Fig. 11 axis; coefficients by least squares on the digitised curve over $1.1 \le \bar{w} - \bar{w}_s \le 5.5$ and the form is used there only; $\bar{w} - \bar{w}_s = 1$ lies outside the band; the gain of the form vanishes at $\bar{w} - \bar{w}_s = -c_0$)

$$[1/1]$$

$$\bar{a}_z(\bar{w} - \bar{w}_s) = \frac{(\bar{w} - \bar{w}_s) + c_0}{(\bar{w} - \bar{w}_s) + d_1}$$

$$\bar{a}_z(1) = \frac{1 + c_0}{1 + d_1}$$

$$\lim_{\bar{w} - \bar{w}_s \to \infty} (\bar{w} - \bar{w}_s)\left(1 - \bar{a}_z\right) = d_1 - c_0$$

$$c_0 = 0.056, \quad d_1 = 1.197, \quad d_1 - c_0 = 1.141$$

$$\bar{a}_z(1) = 0.481, \quad \bar{a}_z(1.1) = 0.503 \text{ (band floor)}, \quad \bar{a}_z = 0 \text{ at } \bar{w} - \bar{w}_s = -0.056$$

$$[2/2]$$

$$\bar{a}_z(\bar{w} - \bar{w}_s) = \frac{(\bar{w} - \bar{w}_s)^2 + c_1\,(\bar{w} - \bar{w}_s) + c_0}{(\bar{w} - \bar{w}_s)^2 + d_1\,(\bar{w} - \bar{w}_s) + d_2}$$

$$\bar{a}_z(1) = \frac{1 + c_1 + c_0}{1 + d_1 + d_2}$$

$$\lim_{\bar{w} - \bar{w}_s \to \infty} (\bar{w} - \bar{w}_s)\left(1 - \bar{a}_z\right) = d_1 - c_1$$

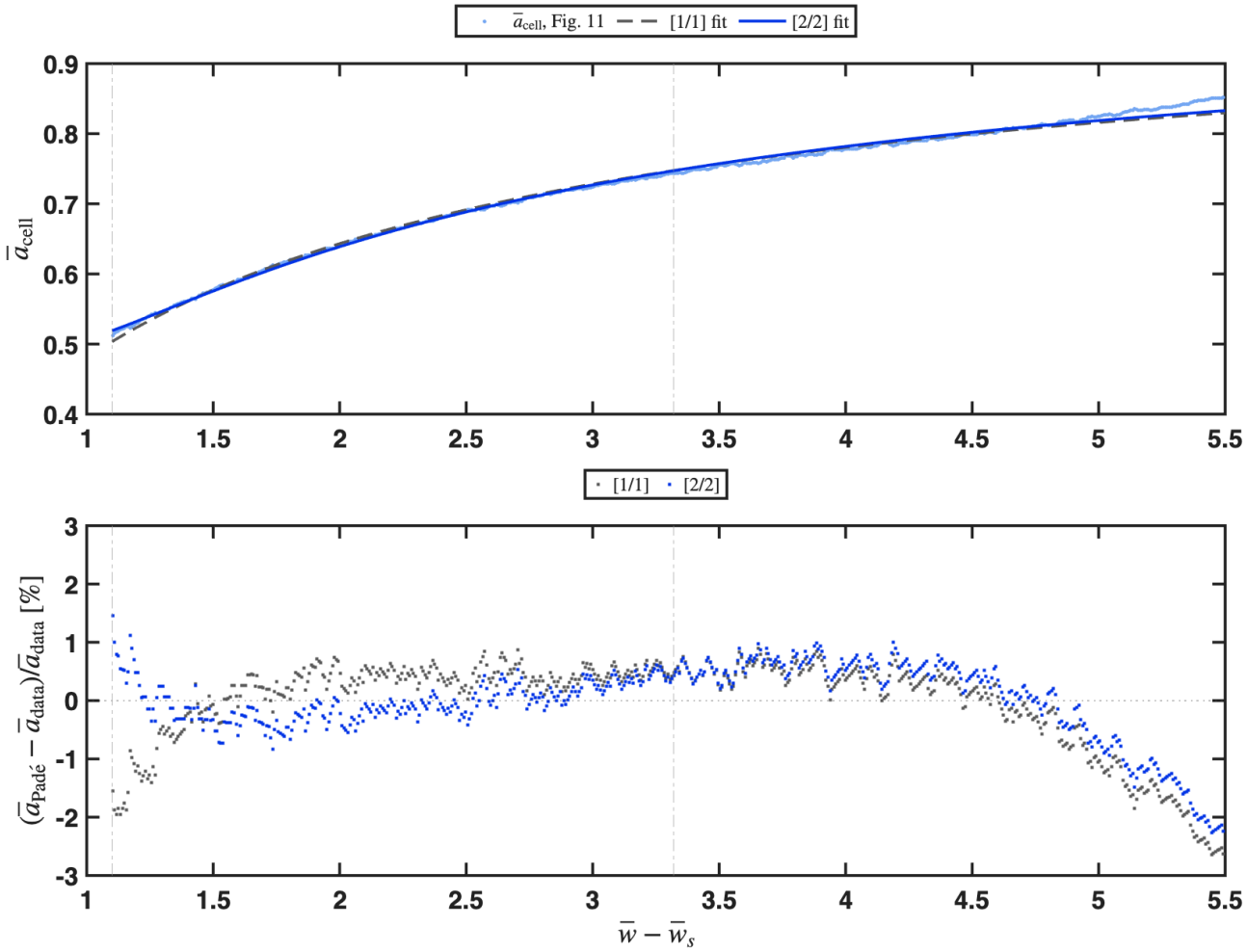
$$c_0 = 0.353, \quad c_1 = -0.571, \quad d_1 = 0.484, \quad d_2 = 0.058$$

$$\bar{a}_z(1) = 0.507, \quad d_1 - c_1 = 1.055$$

same law in the 2026-09-16 coordinate $\bar{w}'_s = \bar{w}_s + 1$ (variable $\bar{w} - \bar{w}'_s = (\bar{w} - \bar{w}_s) - 1$) :

$[1/1]$: $\quad c'_0 = c_0 + 1, \qquad d'_1 = d_1 + 1$

$[2/2]$: $\quad c'_0 = c_0 + c_1 + 1, \qquad c'_1 = c_1 + 2, \qquad d'_1 = d_1 + 2, \qquad d'_2 = d_2 + d_1 + 1$


$\bar{a}_{\text{cell}}$, Fig. 11
[1/1] fit
[2/2] fit
$\bar{a}_{\text{cell}}$
[1/1]
[2/2]
$(\bar{a}_{\text{Padé}} - \bar{a}_{\text{data}})/\bar{a}_{\text{data}}$ [%]
$\bar{w} - \bar{w}_s$